

Important: Text generated by AI and may contain incorrect information, always check the content.

```
```python
Import necessary libraries
import matplotlib.pyplot as plt
import seaborn as sns

Check for the presence of null or categorical values
display(df.isnull().sum())

Understand the distribution of numerical columns
display('Summary statistics for numerical columns:')
display(df[['label', 'confidence', 'social_timestamp']].describe())

Distribution of 'label'
plt.figure(figsize=(8,6))
sns.countplot(x='label', data=df)
plt.title('Distribution of Label')
plt.show()

Distribution of 'confidence'
plt.figure(figsize=(8,6))
sns.histplot(df['confidence'], kde=True)
plt.title('Distribution of Confidence')
plt.show()

Distribution of 'social_timestamp'
plt.figure(figsize=(8,6))
sns.histplot(df['social_timestamp'], kde=True, bins=50)
plt.title('Distribution of Social Timestamp')
plt.show()

Understand the distribution of categorical columns
display('Summary statistics for categorical columns:')
display(df[['subreddit', 'post_id', 'sentence_range', 'text']].describe())

Top subreddits
plt.figure(figsize=(8,6))
sns.countplot(x='subreddit', data=df, order=df['subreddit'].value_counts().iloc[:10].index)
plt.title('Top 10 Subreddits')
plt.xticks(rotation=90)
plt.show()
```



```
Distribution of 'sentence_range'
plt.figure(figsize=(8,6))
sns.countplot(x='sentence_range', data=df, order=df['sentence_range'].value_counts().iloc[:10].index)
plt.title('Top 10 Sentence Ranges')
plt.xticks(rotation=90)
plt.show()

Relationship between 'subreddit' and 'label'
plt.figure(figsize=(8,6))
sns.countplot(x='subreddit', hue='label', data=df)
plt.title('Relationship between Subreddit and Label')
plt.xticks(rotation=90)
plt.show()
```
```

| | 0 |
|------------------|---|
| subreddit | 0 |
| post_id | 0 |
| sentence_range | 0 |
| text | 0 |
| label | 0 |
| confidence | 0 |
| social_timestamp | 0 |

dtype: int64

'Summary statistics for numerical columns:'

| | label | confidence | social_timestamp |
|-------|-------------|-------------|------------------|
| count | 2838.000000 | 2838.000000 | 2.838000e+03 |
| mean | 0.524313 | 0.808972 | 1.518107e+09 |
| std | 0.499497 | 0.177038 | 1.552209e+07 |



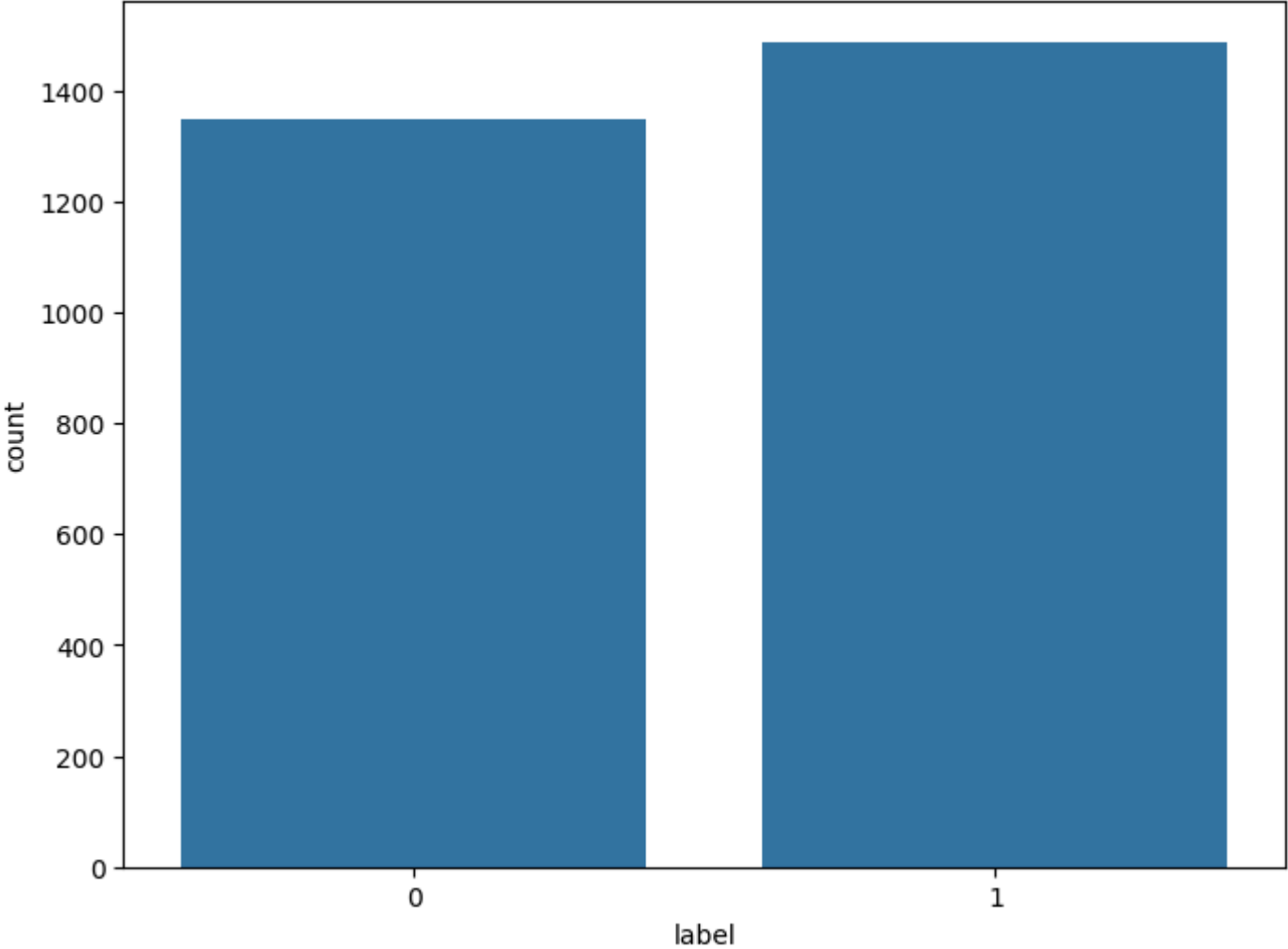


| | label | confidence | social_timestamp |
|-----|----------|------------|------------------|
| min | 0.000000 | 0.428571 | 1.483274e+09 |
| 25% | 0.000000 | 0.600000 | 1.509698e+09 |
| 50% | 1.000000 | 0.800000 | 1.517066e+09 |
| 75% | 1.000000 | 1.000000 | 1.530898e+09 |
| max | 1.000000 | 1.000000 | 1.542592e+09 |



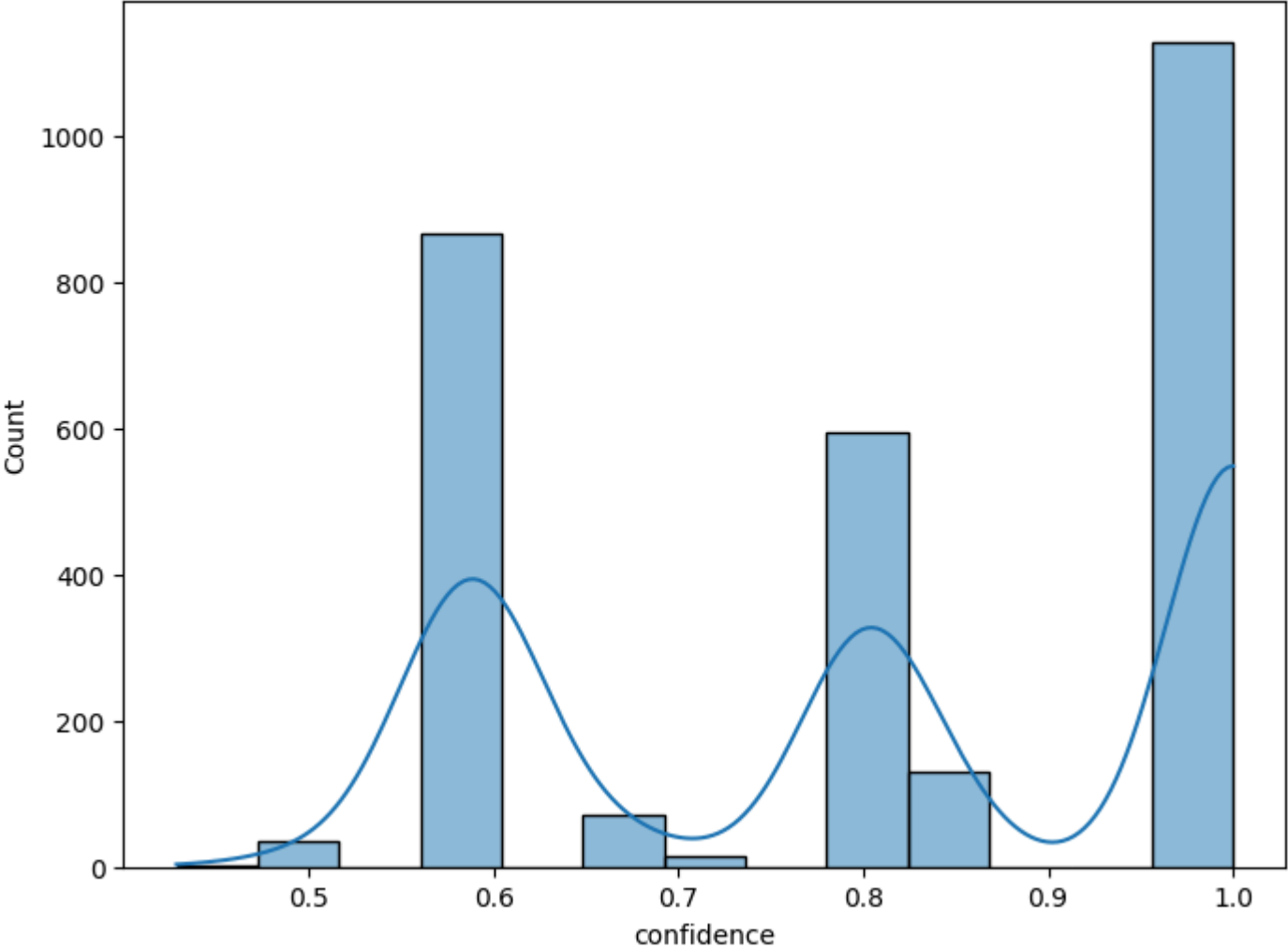


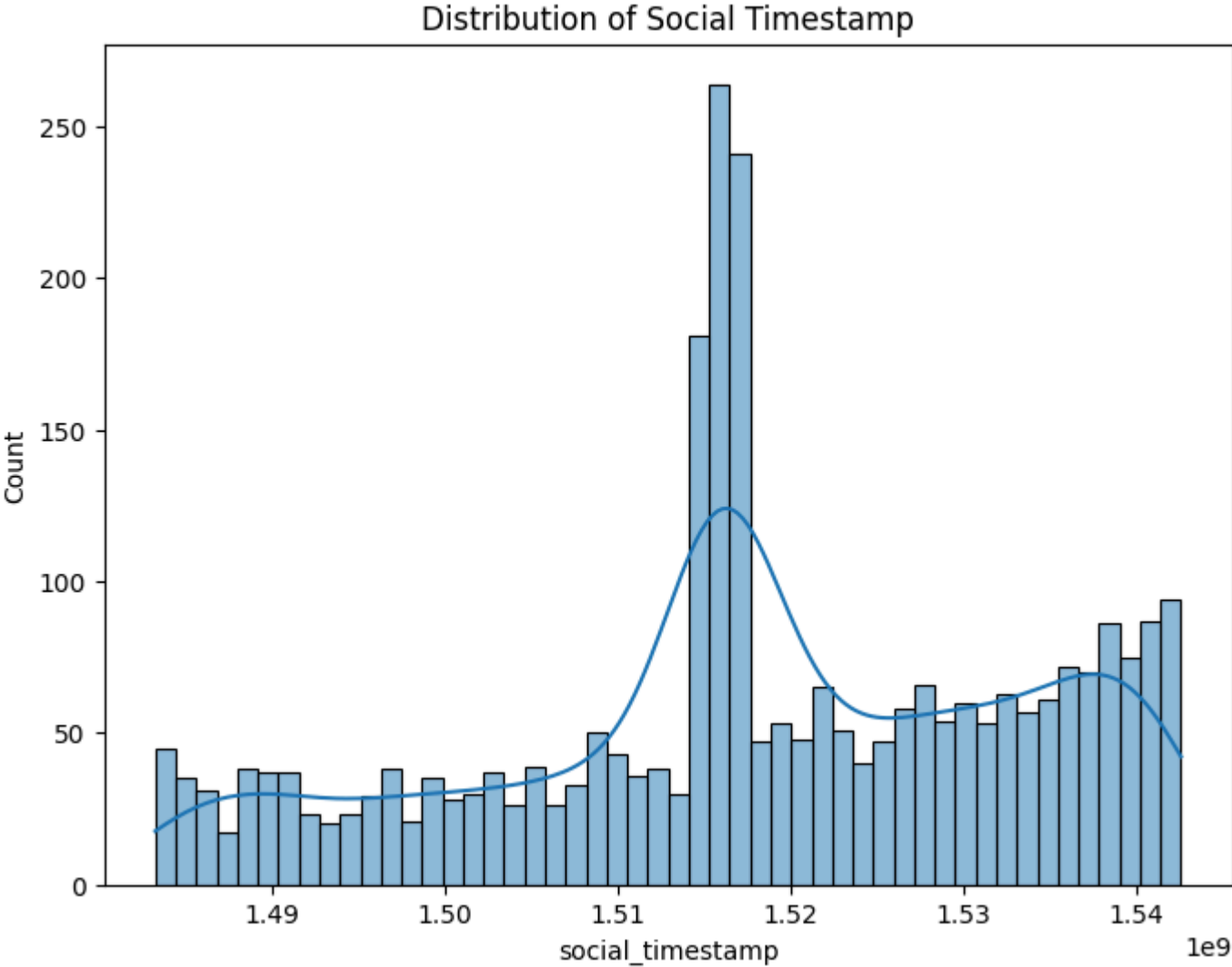
Distribution of Label





Distribution of Confidence





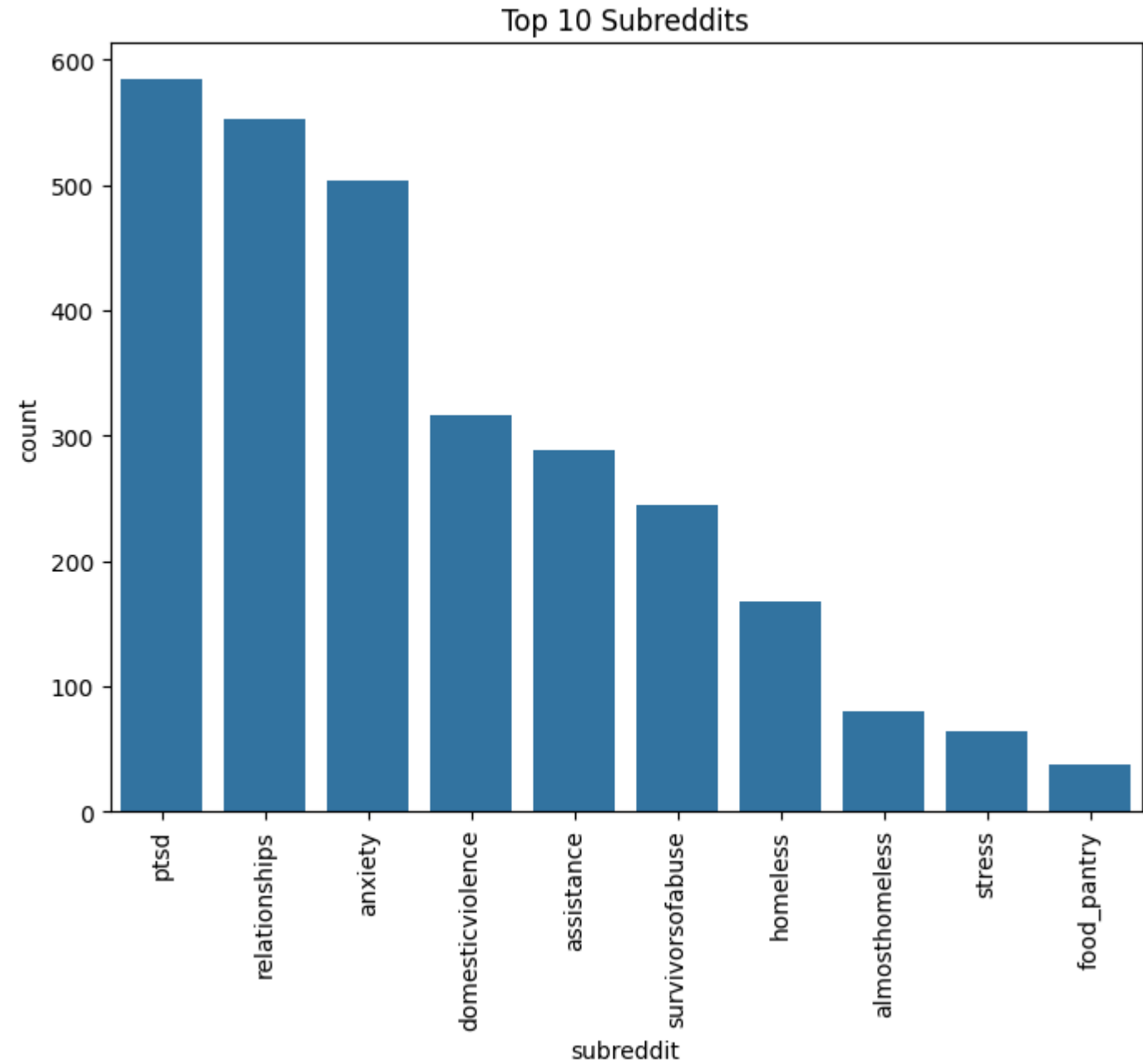
'Summary statistics for categorical columns:'

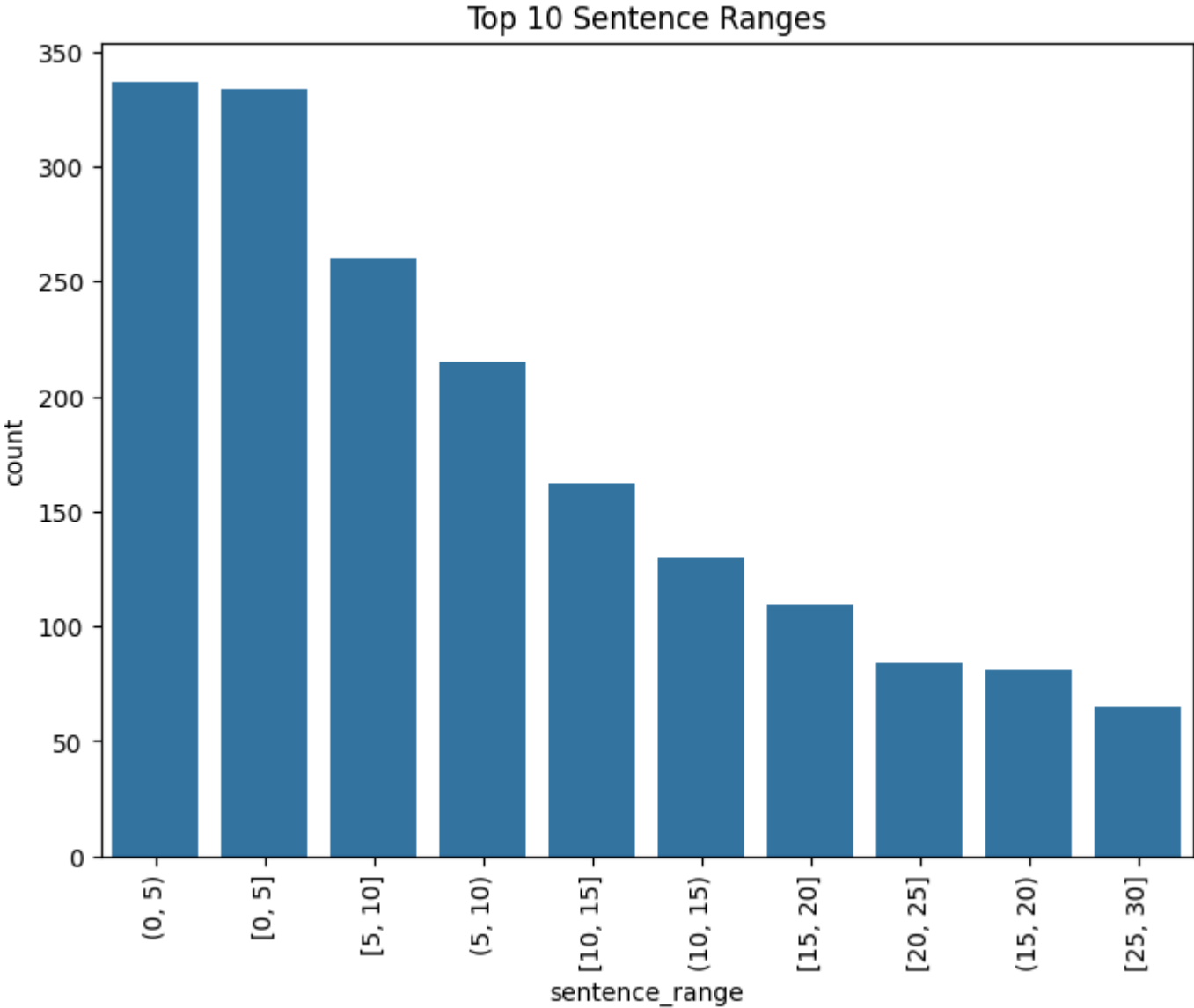
| | subreddit | post_id | sentence_range | text |
|--------|-----------|---------|----------------|---|
| count | 2838 | 2838 | 2838 | 2838 |
| unique | 10 | 2343 | 173 | 2820 |
| top | ptsd | 9md6qz | (0, 5) | Hello, You are invited to complete a survey fo... |





| | subreddit | post_id | sentence_range | text |
|------|-----------|---------|----------------|------|
| freq | 584 | 6 | 337 | 4 |







Relationship between Subreddit and Label

